

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA0305	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	P. Kalyani	Reg. No.:	192472199
Section / Batch:	CSE-AI	Date:	

Code of Conduct

I P. KALYANI (192472199) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **P. Kalyani**

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1

English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.

Apply rules to decompose each word into root and suffix components.

Differentiate between inflectional and derivational forms within the dataset.

Normalize all variants to a common base form for efficient retrieval.

Determine the parsed structure and normalized output for each word.

2. Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.

Identify prefixes, suffixes, and base forms using rule-based decomposition.

Ensure semantic consistency across derived word forms.

Classify each transformation as inflectional or derivational.

Produce the root and morphological breakdown for each word.

3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.

Apply morphological rules to strip affixes and identify the base form.

Handle both grammatical variations and word formation changes.

Ensure consistent root extraction across all forms.

Determine the stem and classification for each input word.

4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.

Apply state transitions to represent suffix transformations and irregular forms in verb inflection.

Differentiate between regular and irregular inflectional patterns within the dataset.
 Ensure consistent mapping of all forms to a common root representation.
 Determine the state transitions, morphological breakdown, and normalized output for each word.

5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.
 Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.
 Handle variations in suffix patterns while preserving semantic consistency.
 Ensure uniform root extraction across all derived forms.
 Determine the stemming steps, intermediate forms, and final stem for each input word.

Rubrics for Calculation

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Morphological Analysis Pipeline Input: connected, connecting, connection

Problem Understanding

The objective is to analyze different forms of the word **connect** by separating the root and suffix, identifying whether the suffix is inflectional or derivational, and normalizing all forms to a common base word for document indexing.

Method Selection

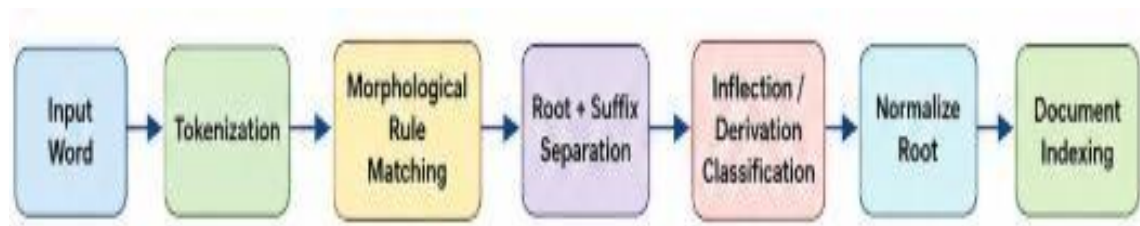
Use a **Rule-Based Morphological Analyzer**. Rules:

- Remove **-ed** → past tense (inflection)
- Remove **-ing** → present participle (inflection)
- Remove **-ion** → noun formation (derivation)

Solution Process

Word	Prefix	Root	Suffix	Type	Normalized Form
connected	—	connect	-ed	Inflectional	connect
connecting	—	connect	-ing	Inflectional	connect
connection	—	connect	-ion	Derivational	conne

Pipeline



Accuracy / Final Result

Word	Parsed Structure	Output
connected	connect + ed	connect
connecting	connect + ing	connect
connection	connect + ion	connect

Interpretation

All words share the same semantic meaning. Normalizing them to connect improves search efficiency because every variation is indexed under one base form.

2. Morphological Parsing for Sentiment Analysis Input: unhappy, happiness, happily Problem Understanding

The task is to decompose words into prefixes, roots, and suffixes while preserving sentiment meaning. The parser should identify derivational changes and map all words to the same base concept.

Method Selection

Use a Rule-Based Morphological Parser.

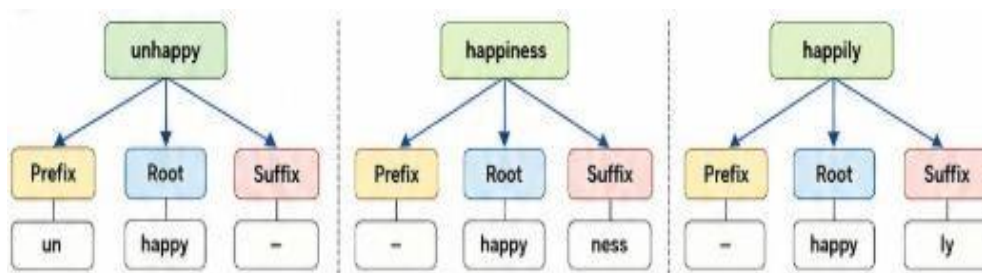
Rules:

- **un-** → negative prefix
- **-ness** → noun suffix
- **-ly** → adverb suffix

Solution Process

Word	Prefix	Root	Suffix	Type	Base Form
unhappy	un	happy	—	Derivational	happy
happiness	—	happy	-ness	Derivational	happy
happily	—	happy	-ly	Derivational	happy

Morphological Breakdown



Accuracy / Final Result

Word	Morphological Structure	Root
unhappy	un + happy	happy
happiness	happy + ness	happy
happily	happy + ly	happy

Interpretation

The parser identifies happy as the semantic base. This helps sentiment analysis treat related words consistently while recognizing polarity changes caused by prefixes.

3. Stemming-Based Preprocessing Input: played, player, playing Problem Understanding

The objective is to remove grammatical and derivational suffixes to extract a consistent stem for search engine indexing.

Method Selection

Apply a Rule-Based Stemmer.

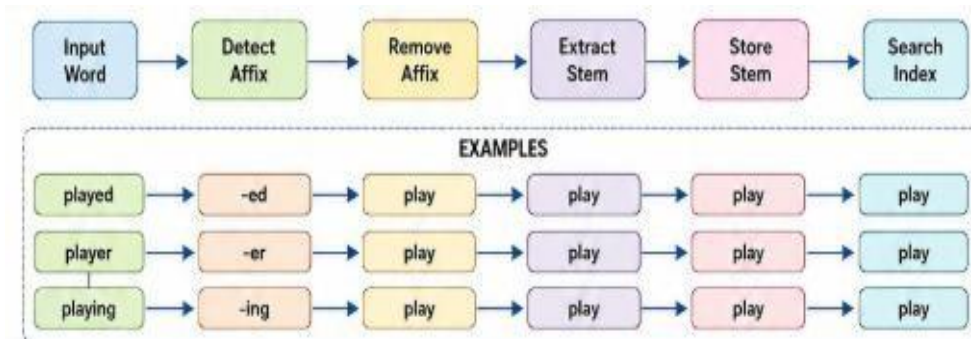
Rules:

- Remove **-ed**
- Remove **-ing**
- Remove **-er**

Solution Process

Word	Root	Removed Suffix	Classification	Stem
played	play	ed	Inflectional	play
player	play	er	Derivational	play
playing	play	ing	Inflectional	play

Processing Flow



Accuracy / Final Result

Word	Stem
played	play
player	play
playing	play

Interpretation

The search engine indexes all three words under play, increasing retrieval accuracy and reducing duplicate entries.

4. Finite-State Morphological Parsing Input: writes, writing, written Problem Understanding

The task is to model verb inflections using a Finite-State Machine (FSM). The parser must identify both regular and irregular verb forms and normalize them.

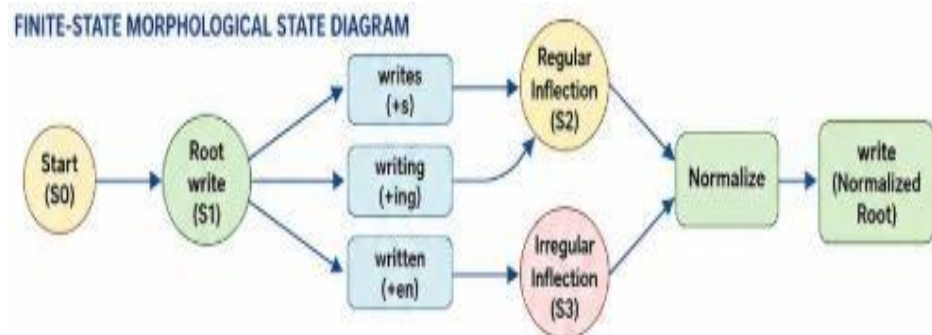
Method Selection

Use a Finite-State Morphological Analyzer.

States:

- S0 → Start
- S1 → Root
- S2 → Regular Inflection
- S3 → Irregular Inflection
- Final → Normalized Root

Solution Process State Diagram



Morphological Table

Word	Breakdown	Pattern	Output
writes	write + s	Regular Inflection	write
writing	write + ing	Regular Inflection	write
written	write + en	Irregular Inflection	write

Accuracy / Final Result

All three words are normalized to:

write

Interpretation

Finite-State Morphology efficiently handles both regular and irregular verb forms while maintaining a common lexical representation.

5. Porter Stemmer-Based Preprocessing Input: relational, relation, relate Problem Understanding

The objective is to apply Porter Stemming step-by-step to reduce derivational word forms into a common stem suitable for document retrieval.

Method Selection

Use the Porter Stemmer Algorithm.

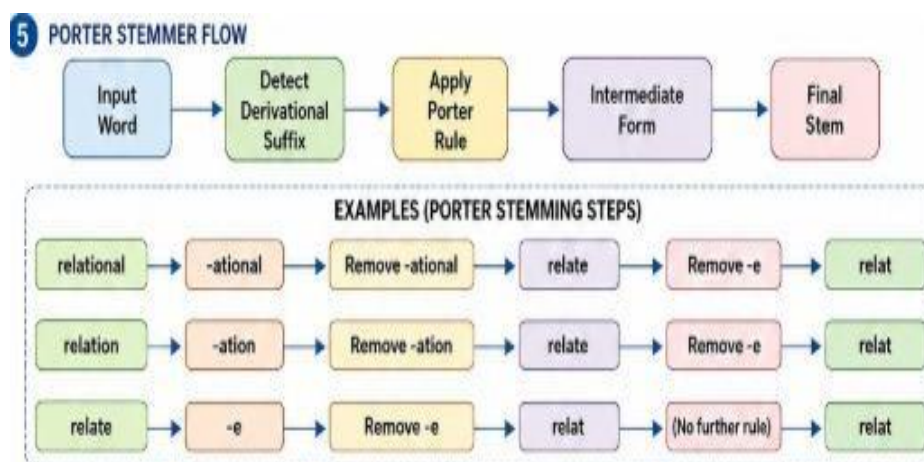
Rules:

- Remove **-ational**
- Remove **-ation**
- Remove **-e**

Solution Process

Word	Step 1	Step 2	Final Stem
relational	relational → relate	relate → relat	relat
relation	relation → relate	relate → relat	relat
relate	relate	remove e	relat

Porter Stemming Flow



Accuracy / Final Result

Word	Intermediate	Final Stem
relational	relate	relat
relation	relate	relat
relate	relate	relat

Interpretation

Porter Stemming maps all related words to the common stem relat, improving indexing performance and enabling efficient document retrieval.